

Introduction to Amazon Web Services

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- Amazon's public cloud
- Full name: Amazon Web Services
- Launched on March 2006
- First service was S3
- ...immediately followed by EC2
- The largest public cloud



- First focused on IaaS services (EC2)
- Later added PaaS
- Currently offers a large variety of cloud services



- Major clients:



Regions

- Amazon built a lot of datacenters for AWS
- Datacenters are grouped in geographic locations
- Each location is called Region
- There are ~40 AWS Regions
- Almost every new resource in the cloud should be allocated to a region

Regions Name

- Regions have a consistent naming scheme
- `geography-direction-serial`
- ie:
- `us-east-1`
- `ap-south-2`

Regions Name

- Examples:

Region	Location
us-east-1	N. Virginia, US
eu-south-1	Milan, Italy
ap-south-1	Mumbai, India
af-south-1	Cape Town, South Africa
ca-west-1	Calgary, Canada

See the full list of regions here:

<https://aws.amazon.com/about-aws/global-infrastructure>



Source: <https://aws.amazon.com/about-aws/global-infrastructure/?p=ngi&loc=1>

Region Selection Consideration

- When selecting a region for your cloud services consider:

Service availability

Not all services are available in all regions

Proximity

Select the region closest to your audience

Price

Services pricing varies between regions

Availability Zones

- Regions have more than one physical datacenter
- Usually three
- Great for availability in case a datacenter fails
- Each datacenter is called Availability Zone
- Some cloud services take advantage of multiple zones in a region

Availability Zones Naming

- Some services are deployed in a specific zone
- Zone name is added as a letter at the end of the region name
- ie:
- `us-east-1a`
- `ap-south-2b`
- Note that the zone-letter assignment is random per account

Global, Regional and Zonal Services

- Cloud services are placed in a region, zone or globally
- Determine their level of isolation and availability
- For example:
 - Regional services can failover between zones in the same region
 - Zonal services depend on their zone, and can access only services in the same zone

Global, Regional and Zonal Services

Global

- IAM – Deployed globally and can be used by resources in all regions
- CloudFront – Deployed globally and can route to resources in any region

Regional

- S3 – Deployed in a region and replicated across zones
- ECS– Deployed in a region and balances load across zones

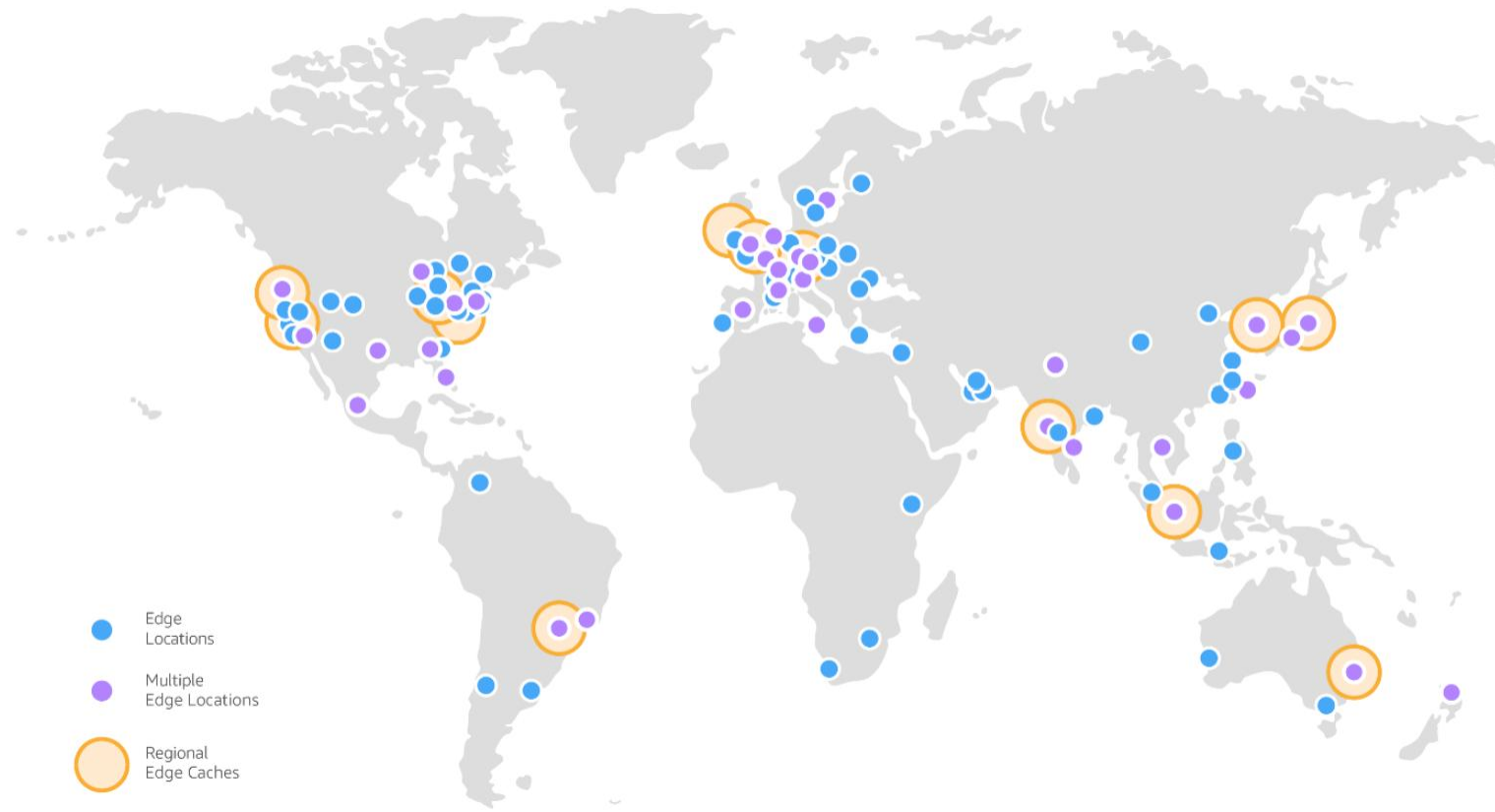
Zonal

- EC2 (VM)– Deployed in a zone and depends on it
- EBS (Disk)– Deployed in a zone

Edge Locations

- AWS locations in addition to the classic regions
- Bring content closer to the end user
- Improve performance
- Distributed worldwide
- More than 400 locations
- Transparent to you

Edge Locations



Edge Locations

- Used by:
 - Content Delivery Network (CDN)
 - DNS servers
 - Lambda@Edge
 - And more...

Local Zones

- AWS datacenters in metro area
- Allow deploying some AWS services closer to end users
- Extension of the parent region
- Should be enabled
- Support networking, compute and data storage
- Connected to parent region with private, high bandwidth network

Local Zones Use Cases

- Low latency applications
 - Gaming
 - Streaming
 - etc...
- Data residency requirements
- Migration of legacy apps

Local Zones

- Generally Available
- Announced



Local Zones

- Supported services:

- EC2

- EBS

- Shield

- ELB

- ECS

- EKS

- VPC

- Direct Connect

- RDS

- And some more...

Resources Hierarchy

- Resources in AWS are organized in hierarchy
- Each resource is contained in its parent resource
 - In most cases can have only a single parent
- Resources inherit policies of their parent

Resources Hierarchy Levels

- Top level of the hierarchy
- Usually represents the company using the cloud
- Have access to all underlying resources
- Defines policies affecting the whole organization
- Optional

Organization

- Groups account
- Useful to apply policies to multiple account
- Should not reflect the organizational tree
- Can contain other OUs
- Optional

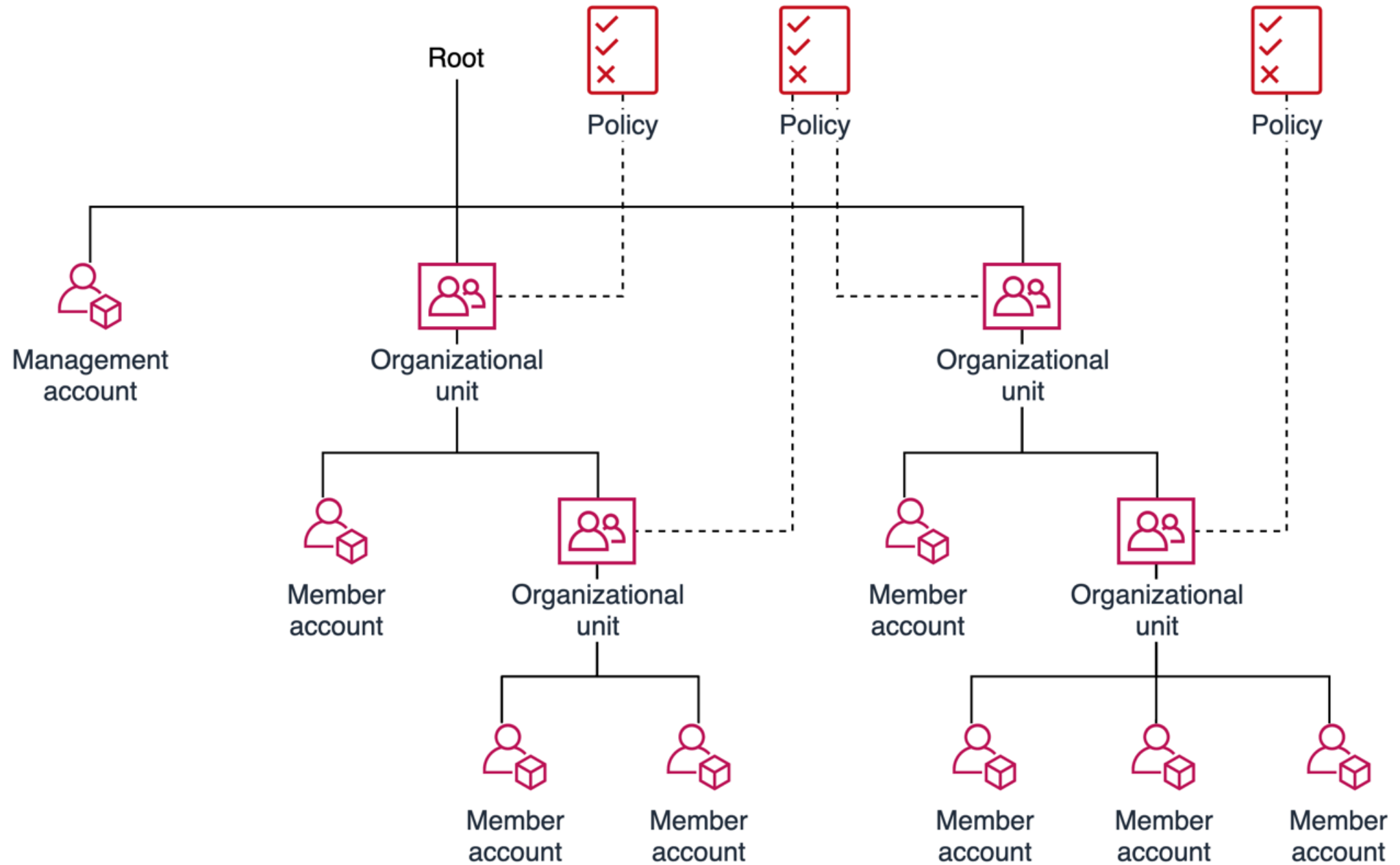
Organizational unit
(OU)

- Contains services
- This is what you'll work with
- Can have exactly one parent OU

Account



Organization



Using Multiple Accounts

- Using multiple accounts allows you to:
 - Group services based on business ownership
 - Apply distinct security controls
 - Manage costs
 - Constrain access to sensitive data
 - And more...

AWS Services

- Everything that is deployed in the cloud is called

Cloud Service

- ie. VMs, databases, networks, AI, user management etc.

AWS Services

- View full list of cloud services in AWS:

<https://aws.amazon.com/>

AWS Free Tier

- AWS offers free tier for new accounts
- The tier is effective immediately and automatically
- Offers free hours / instances / storage
- Usually for a limited amount of time
- Varies per service

AWS Free Tier

Service	Free tier includes...	Valid for...
EC2	750 hours / month on t2.micro or t3.micro instances	12 months
S3	5GB standard storage 20,000 Get requests 2,000 Put requests	12 months
RDS	750 hours / month of db.t2.micro, db.t3.micro, db.t4g.micro instances 20GB of storage	12 months
Load Balancing	750 hours / month 15 LCUs	12 months
API Gateway	1 million API calls / month	12 months